Learned Hyperspectral Image Compression With Coarse-to-Fine Spatial–Spectral Prediction Network

Fanqiang Kong, Qian Li, Pengji Xie, Yunsong Li, *Member, IEEE*, and Pu Tao

***Abstract*—While deep-learned hyperspectral image (HSI) compression has achieved remarkable progress, existing methods typically encode latent representations indiscriminately. This entangled paradigm fails to separate global structural priors from local spectral details, thereby bottlenecking the overall spectral fidelity. To address this issue, we propose a coarse-to-fine spatial-spectral prediction network (CFSP-Net) that strictly decouples HSI reconstruction into two progressive stages: structural estimation and spatial–spectral refinement. Specifically, a lightweight multiscale pooling network first extracts low-frequency priors to provide stable global semantic guidance. Subsequently, a full-resolution fine prediction network (FPN) progressively restores high-frequency local textures and complex spectral correlations via a novel spatial–spectral refinement block, which integrates window-based spatial aggregation and hierarchical cumulative modeling. Furthermore, a fine-grained residual quantization strategy and a band-group optimization scheme are introduced to synergistically stabilize high-dimensional training and refine bit allocation. Extensive experiments on multiple benchmark datasets demonstrate the effectiveness of CFSP-Net, especially on structurally complex HSI scenes and low-to-medium bitrate conditions. Notably, on the Chikusei dataset, the proposed method achieves a peak signal-to-noise ratio (PSNR) of 52.69 dB and a spectral angle mapper (SAM) error of 0.0300 at 0.283 bpppb, improving the PSNR by approximately 0.45 dB over the strongest competing method at a comparable bitrate while maintaining superior spectral fidelity.**

***Index Terms*—Coarse-to-fine compression framework, interspectral prediction, learned hyperspectral image (HSI) compression, residual coding.**

I. Introduction

HYPERSPECTRAL images (HSIs) capture scene information across hundreds of contiguous spectral bands, providing diagnostic physical signatures that are crucial for precise material discrimination in remote sensing applications, such as precision agriculture [1], geological exploration [2], and environmental monitoring [3], [4]. However, the massive data volume resulting from dense spatial–spectral sampling poses severe bottlenecks for storage and transmission. While HSIs exhibit substantial spatial redundancies and strong interband correlations, compressing them is exceptionally challenging. Unlike natural RGB images, HSIs demand extremely strict spectral fidelity, as even minor spectral distortions can completely devastate downstream quantitative analysis and physical retrieval tasks.

Over the past decades, traditional HSI compression algorithms have been extensively investigated, primarily bifurcating into predictive coding and transform-based techniques. Early predictive approaches, such as JPEG-LS [5] and CALIC [6], achieved compression by relying on local spatial correlations. To further exploit the abundant interspectral redundancies, 3-D transform-based algorithms gradually dominated the field, exemplified by 3D-DCT [7], PCA+JPEG2000 [8], 3D-SPIHT, and the extensively deployed CCSDS 123.0-B-2 standard [9], [10]. These conventional codecs benefit from high mathematical stability and low hardware complexity. However, their core mechanisms predominantly depend on hand-crafted rules and linear orthogonal transformations (e.g., the Karhunen–Loéve transform or discrete wavelet transform). Consequently, this reliance on simple linear mathematical models fundamentally restricts their capacity to fully characterize and exploit the highly complex, nonlinear spatial–spectral manifolds inherent in dense hyperspectral data. This limitation inevitably leads to severe performance bottlenecks at low bitrates.

Distinct from traditional linear transform coding, deep-learning-based approaches formulate image compression as a highly nonlinear transform coding problem. By replacing discrete orthogonal transforms with cascades of recurrent or convolutional neural networks (CNNs), these methods enable the joint optimization of all compression modules, including feature transformation, quantization, and entropy coding. This joint optimization explicitly forces the modules to adapt to one another, pushing the rate-distortion (R-D) performance closer to theoretical bounds. Early pioneering works modeled images via binarized latent representations [11], [12]; however, their lack of a trainable entropy model severely restricted rate-control flexibility. To address this, Minnen et al. [13] and Cheng et al. [14] introduced a fully factorized prior with generalized divisive normalization (GDN), and subsequently proposed the Scale Hyperprior. These innovations effectively captured spatial dependencies within the latent representation, establishing the foundational paradigm for modern learned image compression.

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Inspired by the paradigm shift in natural RGB image compression, numerous learning-based frameworks have been specifically tailored for HSIs in recent years [15], [16], [17], [18]. Researchers have explored diverse architectural and statistical strategies to enrich the latent representation and preserve intricate spectral signatures. While early explorations primarily utilized 3D-CNNs to extract joint spatial–spectral features, recent advances have widely introduced hybrid CNN–Transformer architectures to capture long-range global dependencies within the dense data volume. From a statistical perspective, to make more efficient use of allocated bitrates, advanced context models have replaced the standard Gaussian prior with complex probability distributions, such as the Student's t-distribution, to more accurately fit the actual entropy of HSI representations. Furthermore, to mitigate high-frequency information loss during quantization, emerging paradigms like generative adversarial networks (GANs) [19] and cross-channel contrastive learning [20] have been integrated to synthesize realistic spectral details and alleviate spectral distortion.

Despite the remarkable progress of learning-based HSI compression, existing methods still face a critical bottleneck in preserving high-fidelity spatial–spectral information. Most current frameworks improve compression performance by enhancing nonlinear transform coding, entropy modeling, attention-based feature extraction, or residual reconstruction. However, these methods generally encode global structural information and local spectral details within an entangled latent representation. As a result, smooth low-frequency structures and subtle high-frequency spatial–spectral variations are optimized under the same compression bottleneck. During lossy quantization, this coupled representation may lead to spatial blurring, texture attenuation, and spectral drift, especially at low bitrates. Therefore, a more explicit predictive mechanism is needed to separate stable structural estimation from local detail refinement, so that the model can allocate its representation capacity and residual bits more effectively to unpredictable spatial–spectral information.

To address this limitation, we propose a coarse-to-fine spatial–spectral prediction network (CFSP-Net) for learned HSI compression. Different from existing monolithic autoencoder-based compression frameworks, CFSP-Net decomposes the reconstruction process into progressive structural estimation and spatial–spectral refinement. A lightweight coarse prediction network first extracts robust low-frequency representations to provide a stable structural prior. Guided by this prior, a full-resolution fine prediction network (FPN) equipped with a spatial–spectral refinement block (SSRB) restores local high-frequency textures and complex interband correlations. Finally, only the residual between the prediction and the original HSI is encoded, allowing the bitrate to be concentrated on unpredictable spatial–spectral details rather than redundant low-frequency structures. In this way, CFSP-Net provides a prediction-oriented alternative to conventional entangled learned compression paradigms.

The main contributions of this work are summarized as follows.

1) We propose CFSP-Net, a prediction-oriented coarse-to-fine compression framework for HSIs. Unlike conventional monolithic learned compression methods that encode global structures and local spectral details in an entangled latent space, CFSP-Net explicitly separates low-frequency structural estimation from high-frequency spatial–spectral detail refinement, thereby alleviating spatial blurring and spectral distortion under lossy quantization.
2) We design a powerful SSRB within the fine prediction stage. By integrating window-based spatial aggregation and hierarchical cumulative spectral modeling, it effectively captures long-range dependencies and intricate physical spectral correlations.
3) We introduce a decoupled residual coding strategy paired with fine-grained quantization. This mechanism prioritizes precise bit allocation for subtle spectral variations, significantly enhancing the physical fidelity of reconstructed HSIs across diverse bitrates.
4) Extensive experiments on multiple benchmark datasets demonstrate the effectiveness of CFSP-Net, particularly under low-to-medium bitrate conditions and on structurally complex HSI scenes. Notably, on the Chikusei dataset, CFSP-Net achieves a peak signal-to-noise ratio (PSNR) of 52.69 dB and a spectral angle mapper (SAM) error of 0.0300 at 0.283 bpppb, demonstrating strong reconstruction accuracy and spectral fidelity compared with the evaluated traditional and deep learning-based codecs at comparable bitrates.

II. Related Work

In this section, we first review the major developments in deep-learning-based methods for HSI compression. Then, prediction-based image compression strategies are discussed. Finally, we explore the recent advancements of Transformers in low-level vision tasks, which inspired our fine-grained refinement module.

A. Deep-Learning-Based HSI Compression

Deep learning has significantly advanced HSI compression by learning compact latent representations to reduce spatial–spectral redundancies. Fu et al. [21] utilized sparse representation to adaptively capture spatial–spectral correlations. To improve representation capabilities, Sun et al. [17] proposed a CNN–Transformer mixture architecture to capture global dependencies within the dense data volume. Concurrently, Guo et al. [22] introduced an edge-guided framework with interactive dual attention mechanisms to preserve spatial boundaries and texture details during compression. To address the computational burden of processing numerous spectral bands, Verdú et al. [23] investigated a scalable and reduced-complexity deep learning framework to balance R-D performance with computational efficiency. More recently, Wan et al. [24] proposed a super-prior-driven high-fidelity spectral attention network to improve spectral fidelity under constrained bitrates. Wei et al. [25] introduced spectral–spatial

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coupling and group-wise context modeling to better exploit interband redundancy and improve entropy modeling efficiency. Zhou et al. [26] further developed BTC-Net V2 with a spatial-priority hierarchical reconstruction strategy for spaceborne HSI compression. In addition, Liu et al. [27] designed a lightweight band-adaptive compression model with feature decoupling and recurrent band modeling to reduce spatial and spectral redundancy with lower computational overhead. Yu et al. [28] proposed a multiscale wavelet-based spatial–spectral compression network to enhance spectral fidelity and reconstruction quality under low compression ratios.

Furthermore, recent methodologies treat HSIs from entirely different mathematical perspectives. Zhao et al. [29] modeled the spectral dimension as a temporal sequence, exploiting interband redundancies through neural video representation techniques. Alternatively, Rezasoltani and Qureshi [30] combined implicit neural representations (INRs) with coordinate sampling to map spatial–spectral coordinates directly to pixel intensities, providing a continuous, resolution-independent compression strategy. Along this direction, Li et al. [31] introduced low-rank spectral Gaussian splatting to compactly represent HSIs while reducing the computational burden of INR-based compression. In addition, Shi et al. [32] incorporated a Mamba pyramid and spectral difference preservation mechanism to enhance frequency-domain and spectral feature modeling. Additionally, generative models have been integrated into compression pipelines to maintain perceptual quality at low bitrates; for instance, Liu et al. [33] employed a conditional diffusion model within a biresidual network to extract and compress residual information.

Despite these advances, most existing learning-based HSI compression methods still follow an entangled transform-coding paradigm. Hyperprior- and context-model-based compression frameworks mainly improve the statistical modeling of latent representations, but they do not explicitly distinguish low-frequency structural priors from high-frequency spatial–spectral residuals. CNN–Transformer-based HSI compression methods enhance long-range dependency modeling, while Student's t-hyperprior models improve the fitting of latent probability distributions. Other recent methods, such as contrastive-learning-based, INR-based, Gaussian-splatting-based, Mamba-based, and generative residual compression approaches, further improve spectral representation or perceptual reconstruction quality. However, these methods generally focus on strengthening latent representation capacity or reconstruction constraints, rather than explicitly decomposing the HSI reconstruction process into structural prediction and local spatial–spectral refinement. In contrast, CFSP-Net introduces a coarse-to-fine predictive compression framework. The coarse stage provides a stable low-frequency structural prior, and the fine stage progressively models local spatial–spectral variations before residual coding. This design enables more differentiated bit allocation and better preservation of unpredictable high-frequency spectral–spatial details under lossy compression.

B. Prediction-Based Image and Video Compression

Prediction-based compression effectively reduces spatial, temporal, or spectral redundancies by estimating target signals before residual coding. Traditionally, conventional codecs have relied heavily on hand-crafted correlation priors, evolving from pixel-level predictors (e.g., JPEG-LS [5] and CALIC [6]) to sophisticated block-based multimode prediction architectures adopted in HEVC [34] and VVC [35]. Recently, deep neural networks have been deeply integrated into these modules to improve nonlinear estimation accuracy. Dumas et al. [36] proposed a neural network ensemble for intra prediction in the H.265 codec. In video compression, Lu et al. [37] introduced a deep motion compensation network within the DVC framework based on optical flow, while Agustsson et al. [38] proposed scale-space flow to model motion uncertainty. Subsequently, many similar models have been proposed, and improvements have been made to different network components [39], [40], [41], [42]. In the field of remote sensing, Valsesia et al. [43] developed a line-by-line recursive prediction network combining Transformer and RNN structures for onboard HSI compression.

C. Vision Transformers in Compression

While CNNs have historically dominated deep learning, their inherently localized receptive fields severely restrict their capacity to model long-range spatial–spectral dependencies, a critical requirement for processing hundreds of HSI bands. Vision Transformers [44] have emerged as a powerful alternative, demonstrating exceptional capabilities in establishing global contextual priors via self-attention mechanisms. To mitigate the quadratic computational complexity of standard global attention, window-based architectures like SwinIR [45] have been widely adopted in low-level vision to efficiently aggregate high-frequency details. Inspired by this paradigm shift, researchers have actively introduced Transformers into learned image compression [46] to leverage both local and global priors. Rather than merely repurposing generic architectures, our CFSP-Net designs a Transformer-based fine prediction stage equipped with a customized SSRB. This specific design enables the network to explicitly capture intricate physical interband correlations and long-range spatial dependencies, progressively restoring high-frequency textures that pure CNN architectures fail to preserve.

III. Coarse-to-Fine Spatial–Spectral Prediction

In this section, the proposed CFSP-Net, a coarse-to-fine predictive framework for HSI compression, is presented in detail. We first introduce the overall architecture of the framework, and then describe the two key components, including the coarse structural prediction stage and the fine spatial–spectral prediction stage, which collaboratively enable effective residual modeling for compression.

A. Overview

The overall architecture of the proposed method is illustrated in Fig. 1. Unlike conventional end-to-end HSI

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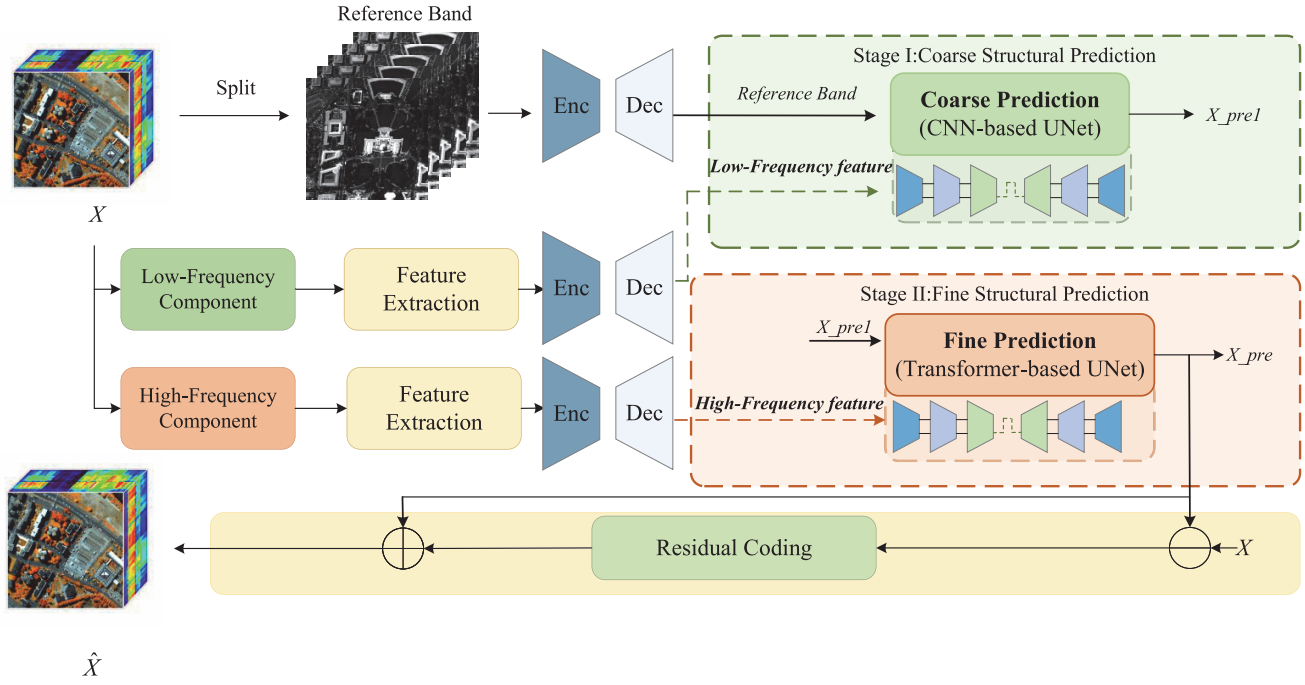


Fig. 1. Overall architecture of the proposed coarse-to-fine spectral prediction network (CFSP-Net) for HSI compression.

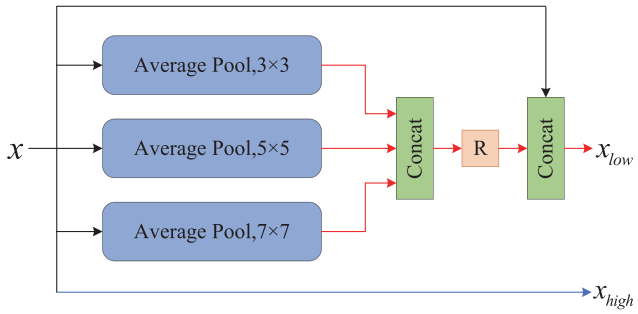


Fig. 2. Frequency decomposition module, where the red and blue arrows denote the low- and high-frequency components, respectively. R denotes the residual block.

compression methods that directly encode latent features, our approach adopts a coarse-to-fine predictive compression framework, which progressively refines the prediction accuracy before residual coding.

The proposed framework consists of three main stages: coarse prediction, fine prediction, and residual compression. The input HSI is first decomposed into groups to capture local spectral correlations. Before the predictive stages, a high-low frequency decomposition module is applied, as illustrated in Fig. 2. Specifically, the original input image is directly retained as the high-frequency component (x_{high}) to completely preserve intricate texture details without any filtering loss. Concurrently, by employing multibranch average pooling (with kernel sizes of 3×3, 5×5, and 7×7) and feature fusion, this module efficiently extracts distinct low-frequency structural components (x_{low}), thereby providing robust multiscale representations. In the coarse stage, a lightweight prediction network is employed to capture the dominant structural information of the target hyperspectral bands. This stage aims to establish a reliable low-frequency and structural prior, which significantly reduces the uncertainty of subsequent prediction. Rather than pursuing complex feature extraction, the coarse prediction focuses on providing a stable and computationally efficient approximation. Subsequently, a Transformer-based Unet fine stage utilizes the coarse prediction as a semantic before guide the precise restoration of high-frequency textures. This stage plays an equally important role as the overall coarse-to-fine framework, as it is responsible for refining local details and spectral correlations across bands. After the two-stage prediction, only the residual between the predicted result and the original HSI is encoded. To further improve compression performance at high bit rates, a fine-grained quantization strategy is applied to the residual representation. This design allows the model to preserve subtle spectral details while avoiding unnecessary bitrate increase, which is particularly important in high-rate scenarios.

B. Coarse-to-Fine Prediction With Spectral–Spatial Refinement

In the top branch, for each 25-band spectral group, five bands with indices $\{0, 5, 10, 15, 20\}$ are uniformly selected as reference bands. This design is motivated by the strong local interband correlation of HSIs, where adjacent or nearby spectral bands usually exhibit higher correlation and more similar material responses. Therefore, the selected reference bands serve as sparse spectral anchors distributed across the group, providing representative spectral information for different local spectral neighborhoods. These anchors help guide the

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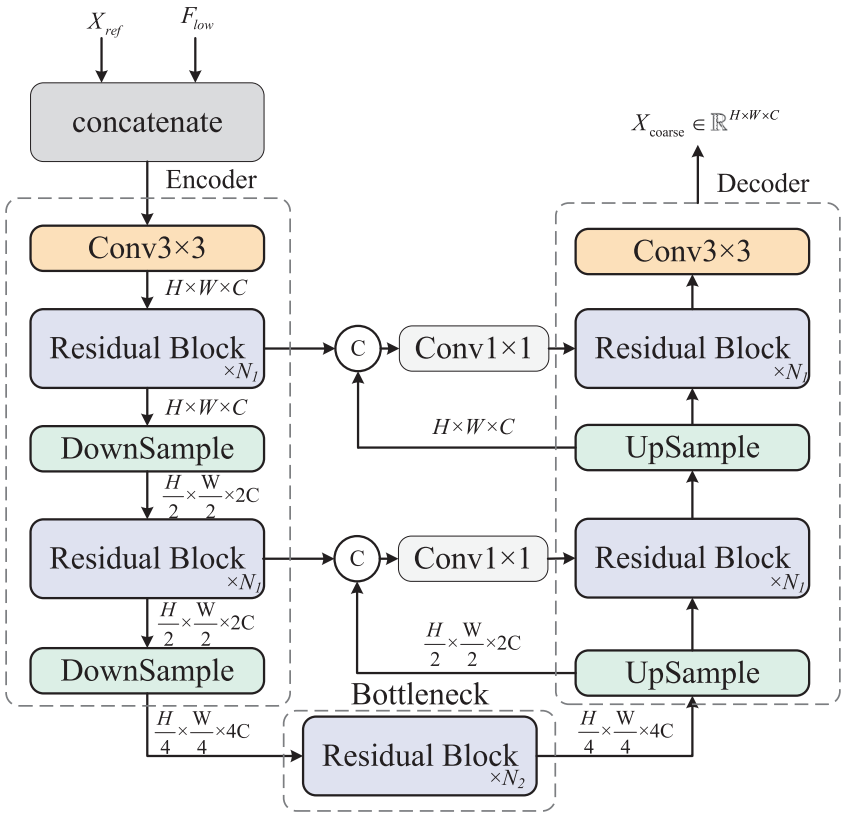


Fig. 3. Architecture of the proposed coarse prediction stage.

prediction of nearby nonreference bands and reduce interband redundancy. The remaining bands are not discarded; instead, they are predicted and reconstructed through the subsequent coarse-to-fine prediction and residual coding stages. It should be noted that this reference-band selection is different from the spatial averaging operation used in the coarse prediction stage. The averaging kernels are applied only along the spatial dimensions to extract low-frequency structural priors and do not determine which spectral bands are retained.

The yellow feature extraction modules in the second and third branches represent learnable neural feature extractors. The second branch extracts coarse low-frequency structural features, while the third branch extracts fine full-resolution spatial–spectral features. Both types of features are learned from the raw HSI data through convolutional layers during end-to-end optimization.

To produce high-quality spectral predictions while maintaining a low bit rate, we propose a coarse-to-fine predictive framework with minimal computation and bandwidth cost at the coarse level. Let $\mathbf{X}^{(g)} \in \mathbb{R}^{C_g\times H\times W}$ denote the current spectral group to be predicted, where C_g, H, and W represent the number of spectral bands, height, and width, respectively. This framework decomposes the complex HSI compression task into two manageable subproblems: global structural estimation and local detail refinement. The coarse level prioritizes the extraction of the low-frequency structural skeleton. Instead of standard downsampling, we employ a multibranch average pooling strategy with varying kernel sizes to suppress high-frequency noise, yielding a robust low-frequency structural representation $\mathbf{F}_{\text{low}}$. The decoded reference feature is denoted as $\mathbf{X}_{\text{ref}}$. To thoroughly leverage contextual dependencies, $\mathbf{F}_{\text{low}}$ is concatenated with $\mathbf{X}_{\text{ref}}$. As illustrated in Fig. 3, this combined representation is then processed by a lightweight CNN-based U-Net to generate a coarse prediction $\mathbf{X}_{\text{coarse}}$, which provides the structural prior for the subsequent fine prediction stage. Specifically, the number of residual blocks at each scale in the encoder and decoder is set to $N_1 = 1$, and $N_2 = 2$ cascaded residual blocks are deployed at the bottleneck. As the fundamental building unit, each residual block consists of two 3×3 convolutional layers paired with ReLU activation functions, followed by a skip connection. This lightweight design allows the network to rapidly establish a stable spatial–spectral prior without imposing significant computational burdens.

While the coarse prediction stage effectively captures the global structural skeleton of the HSI, its simplified modeling and aggressive low-frequency emphasis inevitably lead to the loss of fine spatial textures and subtle spectral variations. To address this limitation, we employ an FPN. Our proposed FPN is illustrated in Fig. 4(a). Inspired by the encoder–decoder paradigm, the FPN is designed to progressively refine local spatial–spectral details while preserving multiscale contextual information. The network operates at the original spatial resolution and adopts a U-shaped architecture with skip connections to effectively balance detail restoration and representation compactness. Specifically, the input feature representation first passes through a sequence of N_1 SSRBs to extract shallow yet detail-sensitive features. This

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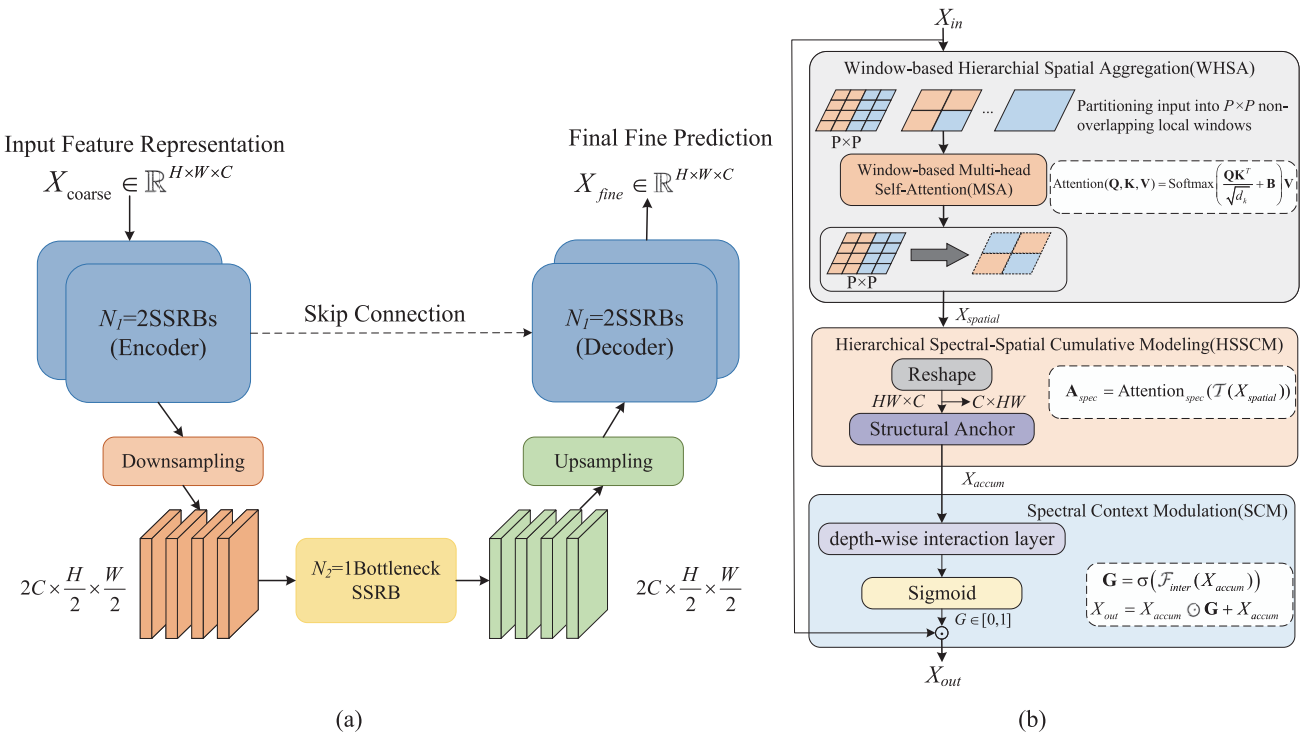


Fig. 4. Architecture of the proposed fine prediction stage. (a) Overall U-shaped architecture of the FPN. (b) Detailed structure of the SSRB.

is followed by a downsampling operation, which reduces the spatial resolution and enables the network to capture higher-level contextual information. The downsampled features are then processed by another set of N_2 SSRBs to further enhance spatial–spectral interactions at a coarser scale. During the decoding stage, features are progressively upsampled and fused with corresponding shallow features via skip connections to recover fine spatial details lost in the coarse stage. After fusion, an additional sequence of N_1 SSRBs is employed to refine the features and produce the final fine prediction. The core unit of the FPN is the SSRB. Unlike standard convolutional blocks that possess limited receptive fields, the SSRB is designed to model long-range dependencies and intricate spectral correlations. As illustrated in Fig. 4(b), the SSRB consists of three cascaded components: Window-based hierarchical spatial aggregation (WHSA), hierarchical spectral–spatial cumulative modeling, and spectral context modulation (SCM).

1) WHSA: Dense spatial textures in high-resolution HSIs are pivotal for maintaining visual fidelity. Although standard global self-attention computes exhaustive pairwise interactions, it suffers from prohibitive quadratic computational complexity. To efficiently capture local fine-grained topological structures without incurring massive computational overhead, we employ a window-based aggregation strategy. Given the input feature map $\mathbf{X}_{in}$, we partition it into nonoverlapping local windows of size $M \times M$. Within each window, pixels are treated as individual tokens to perform multihead self-attention (MSA). For a local window feature $\mathbf{X}$, the query ($\mathbf{Q}$), key ($\mathbf{K}$), and value ($\mathbf{V}$) matrices are generated via linear projections, and the attention matrix is computed as follows:

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{QK}^T}{\sqrt{d}}+\mathbf{B}\right)\mathbf{V} \quad (1)$$

where $\sqrt{d}$ is the scaling factor and $\mathbf{B}$ represents the relative position bias. This mechanism forces the network to concentrate on high-frequency spatial variations within local receptive fields. Furthermore, to enable cross-window interactions and dilate the effective receptive field, a shifted-window partitioning approach is adopted in consecutive blocks, ensuring robust spatial continuity across window boundaries.

2) HSSCM: While the WHSA module effectively captures geometric priors, the resulting features lack explicit spectral coherence. To bridge this gap, the hierarchical spectral–spatial cumulative modeling (HSSCM) module is designed to align spectral dependencies explicitly, utilizing the established spatial structures as deterministic guidance. Specifically, the output of the WHSA module is denoted as $\mathbf{X}_{\text{spa}}$, which serves as a robust structural anchor. We perform a reshaping operation to transpose the spatial tokens into spectral tokens, thereby allowing the attention mechanism to operate along the spectral dimension. The spectrally enhanced feature $\mathbf{X}_{\text{spec}}$ is computed by querying the interband relationships conditioned on the accumulated spatial context

$$\mathbf{X}_{\text{spec}} = \text{SpectralAttention}\left(\mathbf{X}_{\text{spa}}\right). \quad (2)$$

By integrating the structural anchor into the spectral attention calculation, the network learns to adaptively assign continuous weights across bands based on spatial priors. This specific design effectively mitigates spectral aliasing at object boundaries and guarantees that the predicted spectral signatures remain physically and geometrically consistent.

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3) SCM: In the terminal stage of the SSRB, we address the inherent redundancy of the aligned spectral features. Since adjacent hyperspectral bands exhibit extremely high correlation, direct reconstruction may inadvertently amplify noise or lead to redundant bit consumption. To counteract this, the SCM mechanism acts as a dynamic spectral filter. Taking $\mathbf{X}_{\text{spec}}$ as input, SCM employs a lightweight depth-wise convolution $\text{Conv}_{\text{dw}}(\cdot)$ to capture local spectral contexts. The resulting context is subsequently mapped through a nonlinear activation to generate a soft gating mask $\mathbf{M}$

$$\mathbf{M} = \sigma\left(\text{Conv}_{\text{dw}}\left(\mathbf{X}_{\text{spec}}\right)\right) \tag{3}$$

where $\sigma(\cdot)$ denotes the sigmoid function. Finally, the module adaptively recalibrates the spectral responses via element-wise multiplication, followed by a residual connection:

$$\mathbf{X}_{\text{out}} = \mathbf{X}_{\text{spec}} \odot \mathbf{M} + \mathbf{X}_{\text{in}} \tag{4}$$

where $\odot$ represents the Hadamard product and $\mathbf{X}_{\text{out}}$ denotes the final output of the SSRB. The dynamic gate $\mathbf{M}$ suppresses redundant spectral channels while amplifying discriminative absorption features, yielding the finely refined output $\mathbf{X}_{\text{out}}$.

C. Differential Quantization Strategy for Residual Coding

Despite the robust feature extraction and progressive spatial–spectral refinement executed in the prediction stages, an unpredictable residual signal inevitably remains. This residual encapsulates high-frequency geometric textures and subtle spectral discrepancies that elude deterministic prediction. To guarantee ultimate reconstruction fidelity, encoding this residual information is indispensable. Let $\mathbf{X}$ denote the original HSI, and let $\hat{\mathbf{X}}$ represent the final prediction generated by the FPN. The residual signal $\mathbf{R}$ is mathematically formulated as

$$\mathbf{R} = \mathbf{X} - \hat{\mathbf{X}}. \tag{5}$$

We employ a dedicated residual autoencoder to transform $\mathbf{R}$ into a compact latent representation $\mathbf{Y}_{\text{res}}$. To prepare these continuous latents for the subsequent entropy coding, a discrete quantization process is mandatory. While existing variable-rate modulation networks often learn dynamic scalers to cover a wide quality range, our CFSP-Net deliberately adopts a deterministic, hierarchical quantization strategy. This design explicitly enforces the preservation of physical spectral details. The reconstructed (dequantized) latent representation $\hat{\mathbf{Y}}_{\text{res}}$ is derived by

$$\hat{\mathbf{Y}}_{\text{res}} = \Delta_{\text{res}} \cdot \left\lfloor \frac{\mathbf{Y}_{\text{res}}}{\Delta_{\text{res}}} \right\rceil \tag{6}$$

where $\lfloor\cdot\rceil$ denotes the rounding-to-nearest-integer operation, and Δ_{res} represents the quantization step size specifically allocated for the residual features. A paramount design principle within our coarse-to-fine framework is the differential allocation of quantization parameters. For the coarse structural representations, we assign a relatively large quantization step (e.g., Δ_{coarse}) to minimize the bitrate overhead consumed by the low-frequency skeleton. Conversely, for the residual latent $\mathbf{Y}_{\text{res}}$, we strictly enforce a significantly finer quantization step (e.g., $\Delta_{\text{res}} \ll \Delta_{\text{coarse}}$). Rather than relying on a blind, uniform bit allocation paradigm, this decoupled quantization strategy explicitly funnels the majority of the bitrate resources toward the unpredictable residual details. Consequently, it meticulously preserves high-frequency spectral absorption features, substantially elevating the overall physical fidelity of the reconstructed HSI relative to the ground truth.

IV. Experimental Results and Analysis

To rigorously evaluate the compression efficiency and spectral fidelity of the proposed CFSP-Net, extensive experiments were conducted on diverse public hyperspectral datasets. The performance of CFSP-Net is comprehensively benchmarked against both standardized traditional codecs and state-of-the-art (SOTA) deep-learning-based approaches.

A. Datasets

1) Chikusei: Acquired by the Headwall Hyperspec-VNIR-C imaging sensor over agricultural and urban areas in Chikusei, Japan. It comprises 2517 × 2335 pixels with 128 spectral bands ranging from 363 to 1018 nm. A false-color composite is shown in Fig. 5(a).

2) Washington DC Mall: An airborne HSI dataset captured by the Spectral Information Technology Application Center (VA, USA). It features a spatial resolution of 1280 × 307 pixels and 191 spectral bands spanning 400–2400 nm, as depicted in Fig. 5(b).

3) Botswana: Acquired by the NASA EO-1 satellite over the Okavango Delta, Botswana. It has a spatial dimension of 1476 × 256 pixels and 145 spectral bands covering the 400–2500-nm range, with spatial and spectral resolutions of 30 m and 10 nm, respectively, as shown in Fig. 5(c).

4) Pavia Center and University: Both datasets were acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over Pavia, Italy. The Pavia Center dataset has a spatial size of 1096 × 715 pixels with 102 spectral bands, while the Pavia University dataset consists of 610 × 340 pixels with 103 spectral bands. Their false-color composites are presented in Fig. 5(d) and (e).

B. RDO and Training Strategy

To optimize the proposed CFSP-Net, we formulate the training process as a joint R-D optimization (RDO) problem. The overall loss function $\mathcal{L}$ is defined to dynamically balance the reconstruction distortion D and the expected bit rate R

$$\mathcal{L} = D + \lambda R \tag{7}$$

where D denotes the mean-squared error (mse) between the original HSI $\mathbf{X}$ and the final reconstructed image $\hat{\mathbf{X}}$. The hyperparameter λ governs the tradeoff between the physical fidelity and the compression ratio. The total bit rate R aggregates the bit consumption from all submodules, including the coarse structural prior, the intermediate representations, and the final residual. For each component $k \in \{\text{coarse, fine, res}\}$, the bit rate comprises the latent representation $\mathbf{Y}_k$ and its corresponding spatial hyperprior $\mathbf{Z}_k$

$$R = \sum_k \left(\mathbb{E}\left[-\log_2 p_{\hat{\mathbf{Y}}_k|\hat{\mathbf{Z}}_k}\left(\hat{\mathbf{Y}}_k|\hat{\mathbf{Z}}k\right) \right] \right.$$

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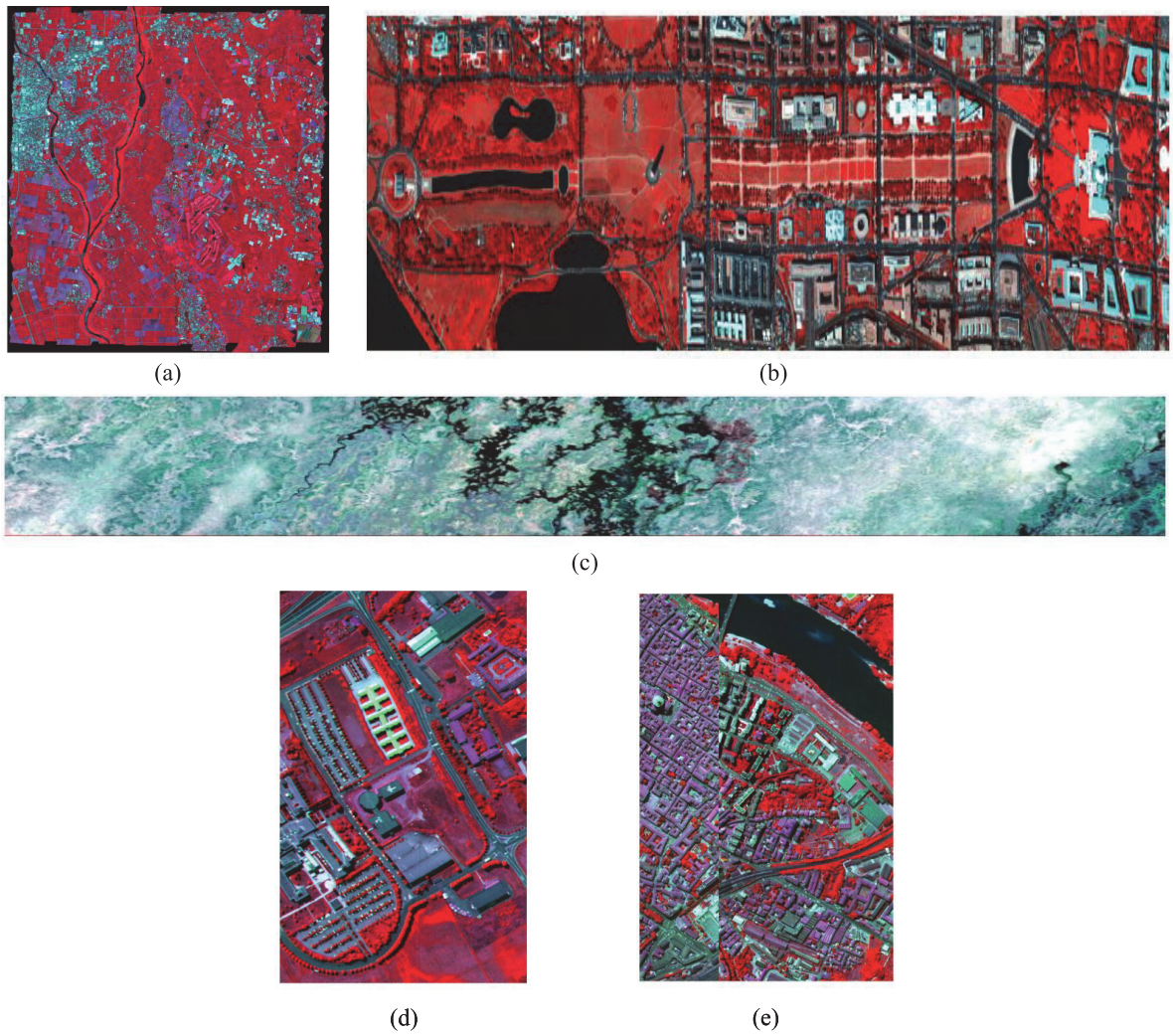


Fig. 5. False-color composites of the evaluated HSI datasets. (a) Chikusei data. The false color is made with bands 100, 10, and 50. (b) Washington DC data. The false color is made with bands 60, 27, and 17. (c) Botswana data. The false color is made with bands 17, 50, and 80. (d) Pavia Center data. The false color is made with bands 100, 10, and 50. (e) Pavia University data. The false color is made with bands 100, 10, and 50.

$$+\mathbb{E}\left[-\log_2 p_{\hat{\mathbf{Z}}_k}\left(\hat{\mathbf{Z}}_k\right)\right]\Bigg). \tag{8}$$

A critical challenge during the end-to-end training is the nondifferentiability of the discrete rounding operation $\lfloor\cdot\rceil$ in the quantization process. To circumvent the zero-gradient issue during backpropagation, we employ continuous uniform noise as a differentiable surrogate for quantization. Crucially, aligning with our proposed differential quantization strategy, the scale of the injected noise is strictly calibrated to the corresponding quantization step size Δ_k. Consequently, the probability density function during training is formulated as the convolution between the underlying continuous distribution and the scaled uniform noise

$$p_{\tilde{\mathbf{Y}}_k|\tilde{\mathbf{Z}}_k} = \left(\mathcal{N}\left(0,\sigma_k^2\right) * \mathcal{U}\left(-\frac{\Delta_k}{2},\frac{\Delta_k}{2}\right)\right)\left(\tilde{\mathbf{Y}}_k\right). \tag{9}$$

This operation physically simulates the differential quantization boundaries, ensuring that the entropy model accurately estimates the bit consumption under varying Δ settings. All models are implemented using the PyTorch framework and trained on a single NVIDIA RTX 4090 GPU. Following the standard learning-based compression paradigm, we utilize two separate Adam optimizers. The primary optimizer updates the main backbone of the three autoencoders with an initial learning rate of 2×10^{-4}, while the auxiliary optimizer refines the parameters of the entropy bottleneck distributions with a fixed learning rate of 1×10^{-3}. To ensure stable convergence, a dynamic learning rate scheduling mechanism is adopted: if the total loss plateaus for 20 consecutive epochs, the primary learning rate is scaled by a factor of 0.2. The training process is terminated when the primary learning rate drops below 1×10^{-7}, which typically occurs within 1000–1500 epochs. To generate the comprehensive R-D curves, models are trained at distinct operational points by varying the λ values (e.g., from 5×10^{-5} to 0.0015), enabling flexible control over the reconstruction quality across diverse datasets.

C. Evaluation Metrics

To comprehensively evaluate the compression efficiency and reconstruction fidelity of the proposed CFSP-Net, we employ bits per pixel per band (bpppb) to quantify the compression

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ratio. Furthermore, three widely adopted distortion metrics are utilized to rigorously assess spatial and spectral quality: PSNR, multiscale structural similarity (MS-SSIM) [47], and SAM [48].

1) PSNR: The PSNR serves as a fundamental indicator of spatial reconstruction accuracy, where a higher value implies less distortion. Considering the distinct radiometric characteristics across different hyperspectral bands, calculating the PSNR using a globally fixed peak value is highly suboptimal. Therefore, we compute the peak signal independently for each individual band. For an HSI comprising C spectral bands, the overall PSNR is defined as the mean across all channels

$$\mathrm{PSNR} = \frac{1}{C}\sum_{i=1}^{C} 10\lg\left(\frac{(\mathrm{MAX}_i)^2}{\mathrm{MSE}_i}\right) \tag{10}$$

where MAX_i denotes the maximum possible pixel value in the ith band, and mse_i is the mse of that specific band.

2) MS-SSIM: The MS-SSIM evaluates the visual consistency and structural preservation between the original and decompressed images across various scale levels. To better illustrate minor performance discrepancies at high-quality regimes, we convert the raw MS-SSIM scores into a logarithmic decibel format

$$\text{MS-SSIM} = -10\lg(1 - e_{\text{MS-SSIM}}) \tag{11}$$

where $e_{\text{MS-SSIM}}$ denotes the raw MS-SSIM measurement computed between the original and reconstructed HSIs.

3) SAM: While the PSNR and MS-SSIM focus on spatial fidelity, the SAM explicitly quantifies the physical spectral consistency. It calculates the spectral angle between the original and reconstructed hyperspectral pixels.

A lower SAM value strictly indicates less spectral drift. The global SAM metric is defined as

$$\mathrm{SAM}\left(X, \hat{X}\right) = \frac{1}{H \times W} \times \sum_{j=1}^{H\times W} \cos^{-1}\left(\frac{x_j \odot \hat{x}_j}{|x_j||\hat{x}_j|}\right) \tag{12}$$

where X and $\hat{X}$ represent the original and reconstructed HSIs, respectively, and $H \times W$ denotes spatial resolution. The vectors x_j and $\hat{x}_j$ correspond to the spectral signatures of the jth pixel, with $\odot$ denoting the inner product of the two spectral vectors, and $\|\cdot\|$ representing the vector norm.

D. Implementation Details and Baseline Methods

To ensure fair and consistent evaluation across datasets with different spatial sizes and spectral band numbers, we reprocessed all datasets using a unified patch generation protocol. Each hyperspectral cube was normalized to 16-bit unsigned integers and cropped into 256 × 256 spatial patches. For training, patches were extracted with a stride of 128 to increase the number and diversity of training samples. For testing, patches were extracted with a stride of 256 to reduce overlap among test samples. The testing regions were first selected on a dataset-wise spatial grid, and any training patch overlapping with the selected testing regions was removed to avoid information leakage. Along the spectral dimension, each sample contains 50 contiguous bands, and adjacent spectral windows were generated with a stride of 25. The last valid spectral window was also included to preserve complete spectral coverage. Therefore, the spectral continuity of HSIs was maintained during preprocessing.

To rigorously benchmark the compression efficiency and reconstruction fidelity of the proposed CFSP-Net, we compared it against two distinct categories of compression algorithms. The first category comprises representative traditional codecs: JPEG2000 (implemented via the official OpenJPEG library), PCA+JPEG2000 [8], and 3D-SPIHT [49]. The second category encompasses SOTA deep learning-based models, including the generic Scale Hyperprior [12] and Joint Autoregressive Context [13] frameworks. Additionally, to evaluate performance against domain-specific neural architectures, we included FDRCN [50], a high-performance deep residual convolutional network specifically tailored for multispectral image compression, as a critical baseline.

All deep-learning-based architectures, including our CFSP-Net, were implemented using the PyTorch framework. The networks were optimized using the Adam optimizer, and the initialization and decay scheduling of the learning rate strictly followed the strategy detailed in [12]. During the training phase, data enhancement techniques—including random cropping, horizontal and vertical flipping, and random rotation—were applied to the extracted patches to prevent overfitting and enhance spatial–spectral generalization.

In addition, to clarify the parameter tuning strategy, all learning-based models were trained under unified settings. For the proposed CFSP-Net, the network architecture parameters, including the feature channels, prediction modules, residual coding structure, and quantization strategy, were kept fixed across all datasets. The rate-control parameter λ was only used to generate different bitrate points and was not tuned for any specific test dataset. During inference, the trained model was directly applied to each test set without modifying the network structure or adjusting model parameters. Therefore, the reported results are obtained under a unified compression setting rather than dataset-dependent parameter optimization.

To further evaluate the generalization ability of the proposed method, we additionally constructed unseen cross-dataset test sets using KSC and Houston University data, which were not included in the training set. These hyperspectral cubes were cropped into 256 × 256 patches with continuous 50-band spectral windows following the same testing protocol. For large-scale scenes, only representative nonoverlapping spatial regions were selected to avoid excessive redundancy and computational burden. The trained CFSP-Net was directly tested on these unseen data without retraining or fine-tuning. This setting provides a more direct evaluation of cross-dataset generalization and helps verify that the proposed method learns general spatial–spectral redundancy modeling rather than relying on dataset-specific parameter tuning.

Model training was accelerated on an NVIDIA GeForce RTX 4090 GPU. To ensure a strictly fair comparison regarding computational complexity and inference efficiency, the decoding phases for all evaluated codecs (both traditional

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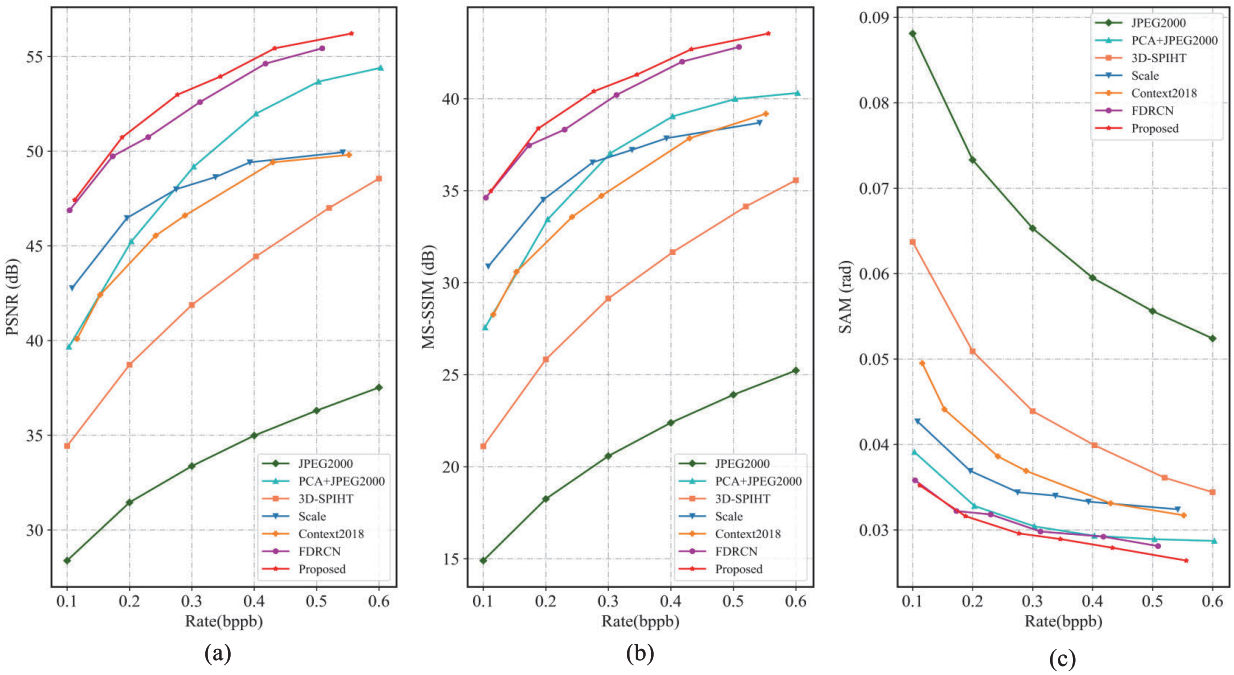


Fig. 6. Rate–distortion curves from the PSNR, MS-SSIM, and SAM perspectives on the Chikusei dataset. (a) PSNRs. (b) MS-SSIMs. (c) SAMs. The proposed method (red line) achieves the best RD performance on these three metrics compared with other methods.

and learning-based) were executed on a single Intel Core i9-10850K CPU running at 3.60 GHz.

E. Experimental Results

1) Compression Performance Comparisons: In this section, we compare the proposed CFSP-Net with representative traditional and learning-based compression methods. The compared methods include three traditional codecs, that is, JPEG2000, 3D-SPIHT, and PCA+JPEG2000, two widely used learned image compression models, that is, Scale Hyperprior and Joint Autoregressive Context, and a specialized multispectral compression model, that is, FDRCN. The experiments are conducted on five hyperspectral test scenes, including Chikusei, Washington DC Mall, Pavia Center, Pavia University, and Botswana. Since Pavia Center and Pavia University share similar urban land-cover characteristics, their results are reported together as the Pavia group.

Fig. 6 shows the R-D curves on the Chikusei dataset in terms of the PSNR, MS-SSIM, and SAM. Chikusei contains complex agricultural fields, urban structures, and dense high-frequency spatial textures, making it challenging for both traditional transform-based codecs and learning-based compression models. As shown in Fig. 6(a) and (b), CFSP-Net achieves the best PSNR and MS-SSIM performance across almost all bitrates. Compared with FDRCN, which is the strongest learning-based baseline, the proposed method provides consistently higher reconstruction fidelity, especially in the medium- and high-bitrate ranges. In terms of the SAM, CFSP-Net also obtains the lowest spectral angular distortion among all compared methods, demonstrating its ability to preserve spectral consistency while improving spatial reconstruction quality. These results indicate that the coarse-to-fine prediction framework is particularly effective for scenes with rich local spatial–spectral variations.

The visual comparison in Fig. 7 further confirms this observation. Under low-bitrate compression, traditional codecs and conventional learned compression models tend to blur field boundaries and weaken local textures. By contrast, CFSP-Net first estimates a low-frequency structural prior and then refines local spatial–spectral residuals through the fine prediction stage. As a result, the reconstructed regions retain clearer object boundaries and more distinguishable texture details.

Fig. 8 presents the R-D curves on the Washington DC Mall dataset. The proposed method achieves competitive PSNR and MS-SSIM performance and clearly outperforms the general learned compression baselines, including Scale Hyperprior and Joint Autoregressive Context. At low and medium bitrates, CFSP-Net achieves strong spatial reconstruction quality, while at high bitrates, PCA+JPEG2000 becomes highly competitive and slightly outperforms the proposed method in the PSNR. A similar phenomenon can be observed in the SAM, where PCA+JPEG2000 achieves the lowest spectral angular distortion. This result is related to the strong global spectral correlation and relatively regular structures in the Washington DC scene. Under high-bitrate conditions, PCA-based global orthogonal decorrelation can effectively preserve the dominant spectral energy and spectral direction. Nevertheless, CFSP-Net still achieves the best or near-best performance among learning-based methods and maintains strong spatial–spectral reconstruction ability. The visual comparisons in Fig. 9 further show that CFSP-Net preserves clearer spatial structures and more faithful spectral information than the competing methods at similar bitrates on the Washington DC Mall dataset.

Figs. 10 and 11 report the quantitative and visual results on the Pavia group. The Pavia scenes contain dense urban structures, complex roof textures, road regions, and fragmented shadow boundaries, imposing high requirements on local detail preservation. As shown in Fig. 10(a) and (b), CFSP-Net

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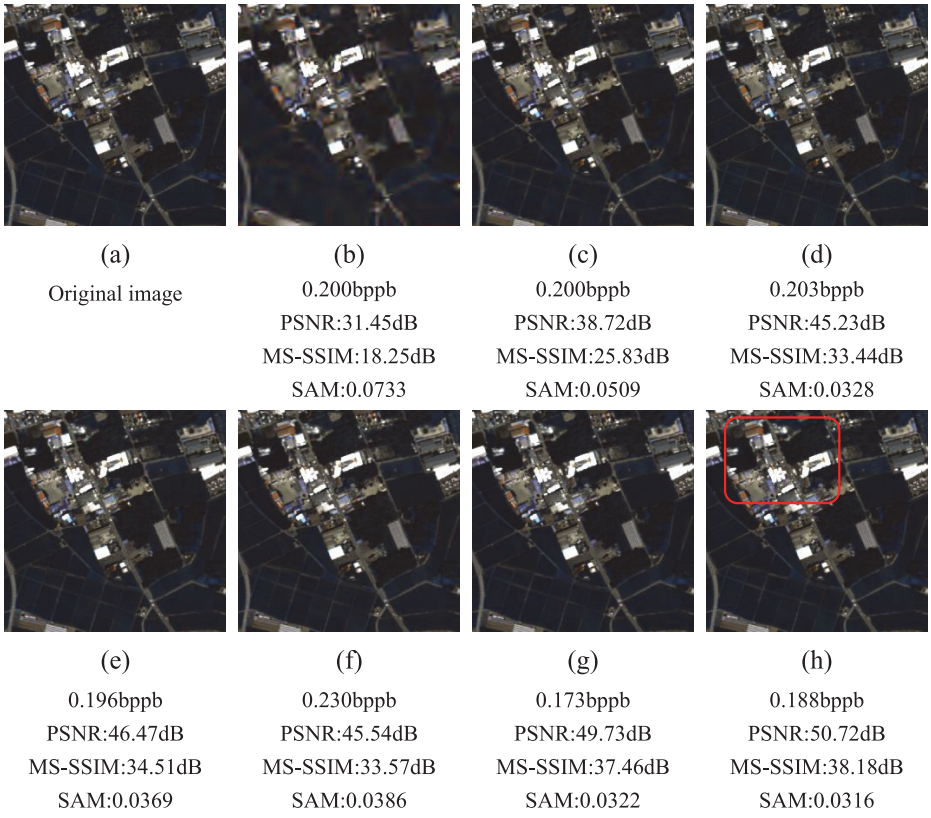


Fig. 7. Visual comparisons at close bit rates on an example patch of the Chikusei dataset. Bands 30, 20, and 10 were used to create the false color. (a) Original image. (b) JPEG2000. (c) 3-D-Spiht. (d) PCA+JPEG2000. (e) Scale. (f) Context. (g) FDRCN. (h) Proposed.

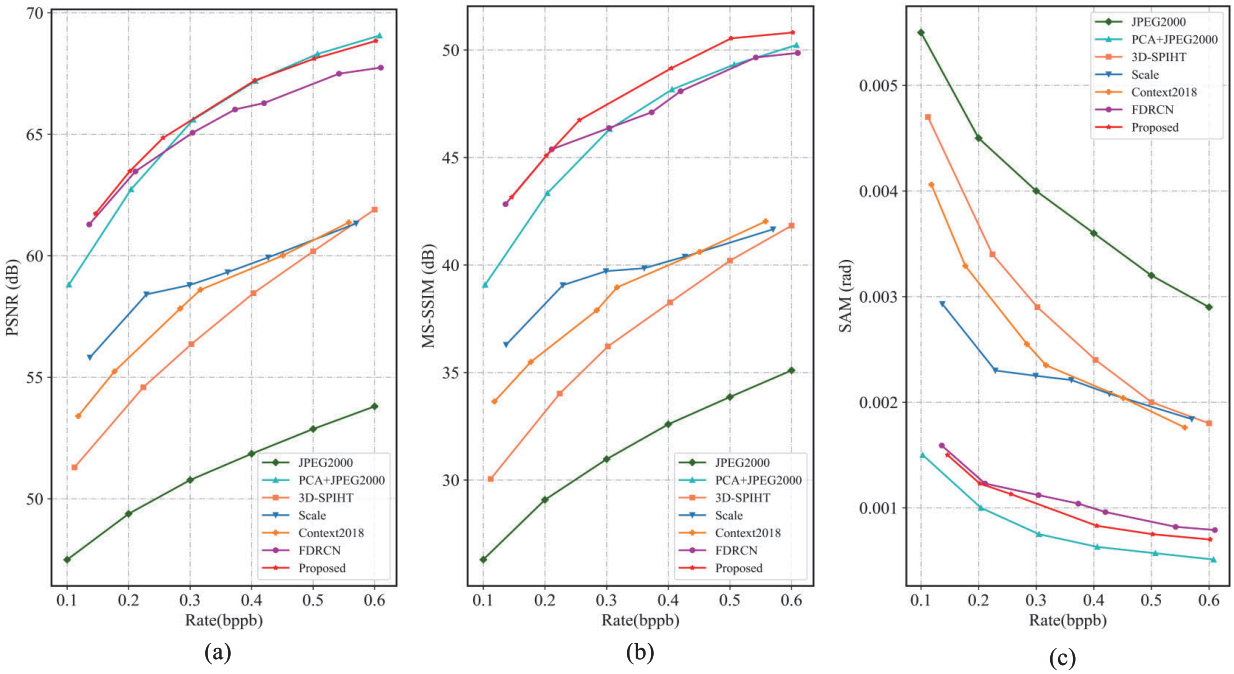


Fig. 8. Rate–distortion curves from the PSNR, MS-SSIM, and SAM perspectives on the Washington DC dataset. (a) PSNRs. (b) MS-SSIMs. (c) SAMs.

achieves the best PSNR and MS-SSIM performance in most bitrate ranges. Compared with Scale Hyperprior, Joint Autoregressive Context, and FDRCN, the proposed method better preserves structural similarity and spatial fidelity. In terms of SAM, CFSP-Net achieves competitive spectral fidelity and outperforms most learning-based baselines, although PCA+JPEG2000 obtains slightly lower SAM values at some medium and high bitrate points. Overall, CFSP-Net provides a better

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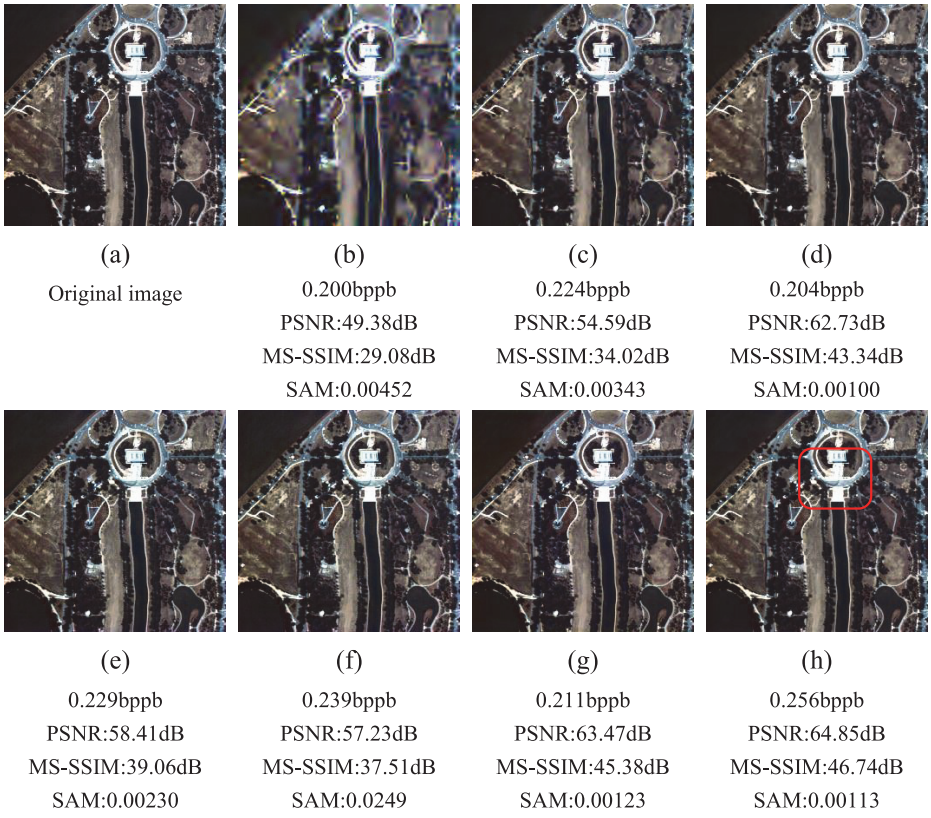


Fig. 9. Visual comparisons at close bit rates on an example patch of the Washington DC dataset. Bands 30, 20, and 10 make up the false color. (a) Original image. (b) JPEG2000. (c) 3D-SPIHT. (d) PCA+JPEG2000. (e) Scale. (f) Context. (g) FDRCN. (h) Proposed.

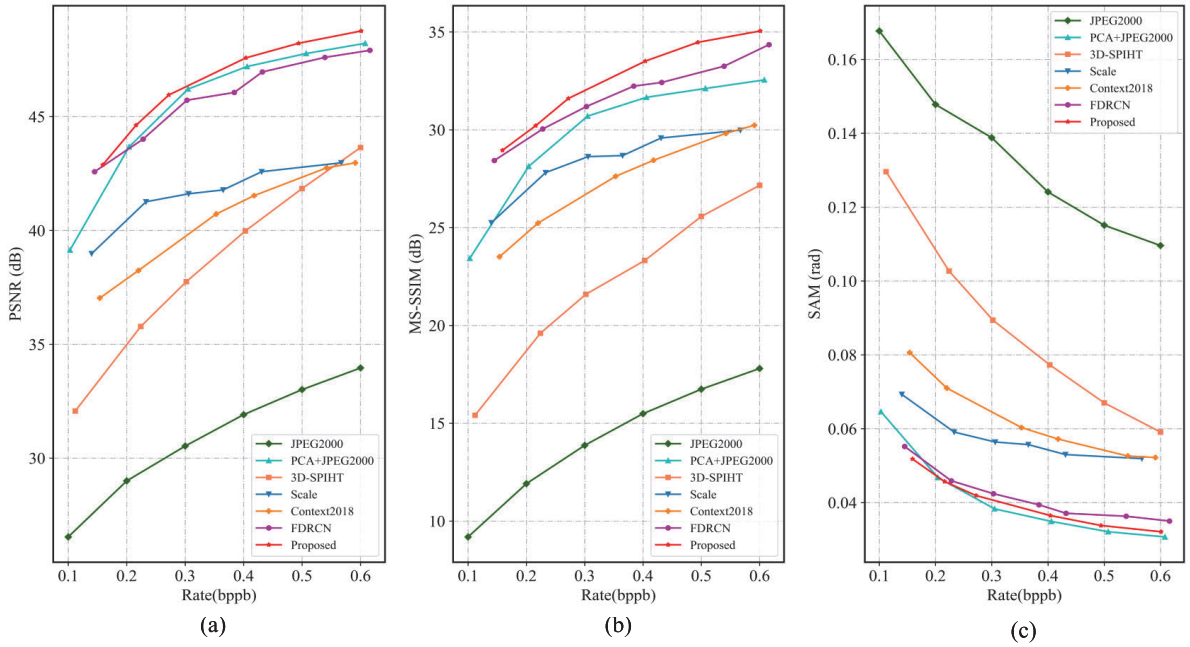


Fig. 10. Rate–distortion curves from the PSNR, MS-SSIM, and SAM perspectives on the Pavia dataset. (a) PSNRs. (b) MS-SSIMs. (c) SAMs.

balance between spatial reconstruction quality and spectral consistency on urban HSI scenes.

Finally, Fig. 12 shows the R-D performance on the Botswana dataset, and Fig. 13 provides the corresponding visual comparisons. Botswana is dominated by natural landscapes, such as vegetation and water bodies, with relatively smooth spatial structures and strong global spectral correlation. On this dataset, CFSP-Net achieves the best PSNR and MS-SSIM performance across most bitrates, indicating that the proposed framework can effectively reconstruct both

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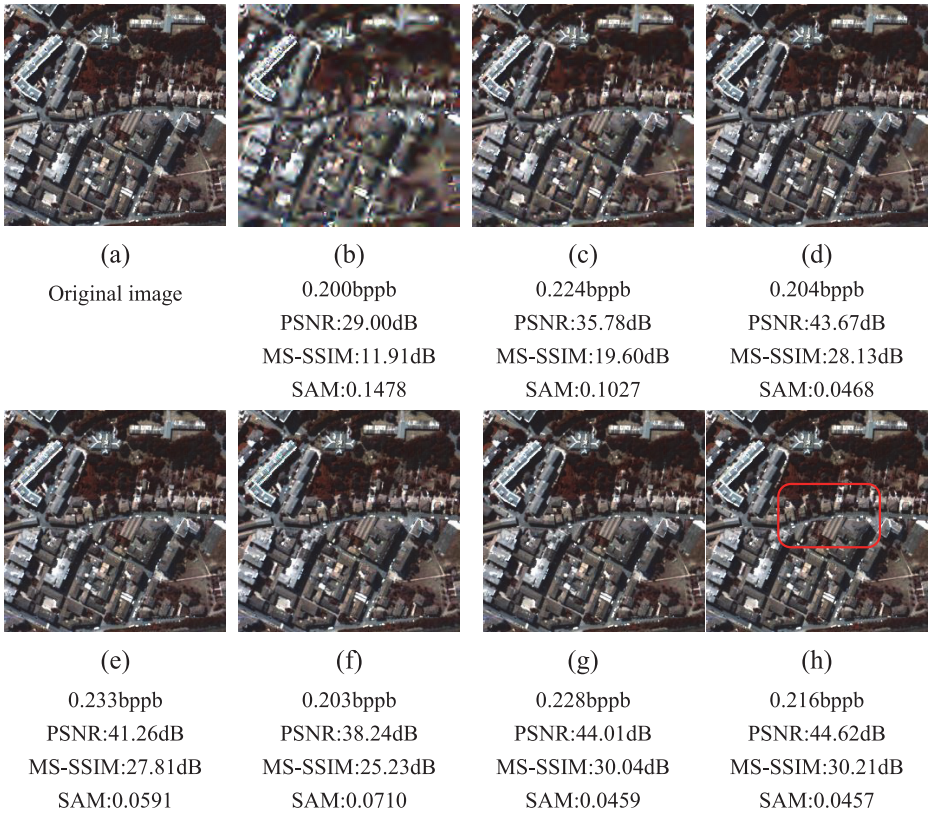


Fig. 11. Visual comparisons at close bit rates on an example patch of the Pavia dataset. Bands 30, 20, and 10 make up the false color. (a) Original image. (b) JPEG2000. (c) 3-D-Spiht. (d) PCA+JPEG2000. (e) Scale. (f) Context. (g) FDRCN. (h) Proposed.

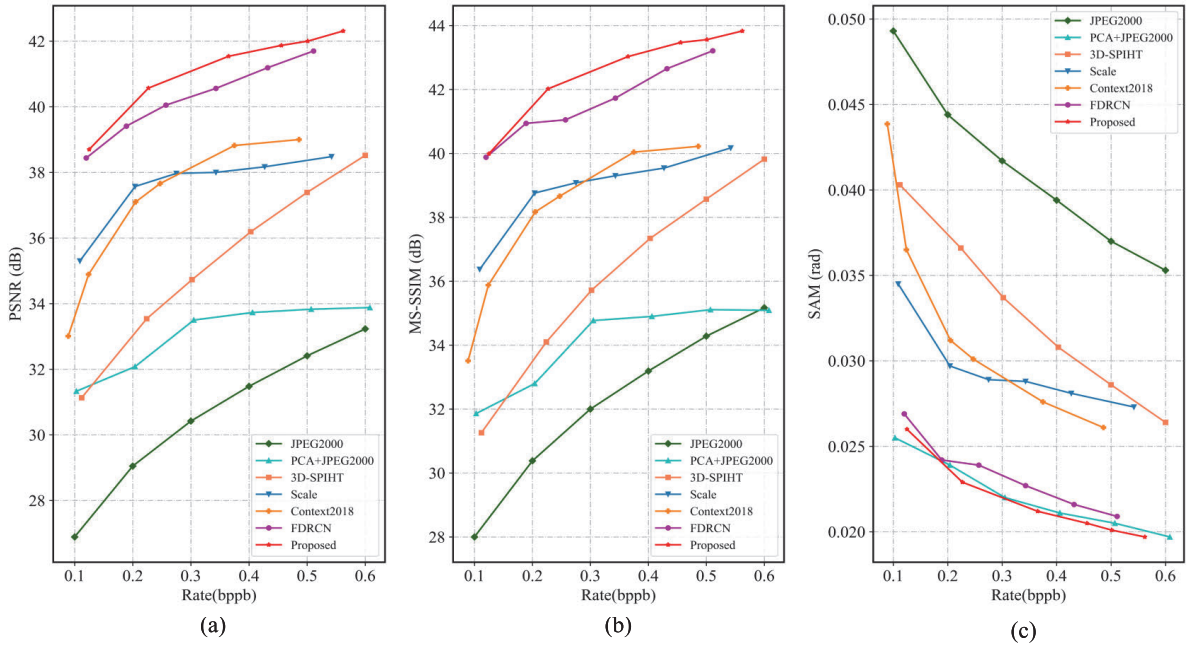


Fig. 12. Rate–distortion curves from the PSNR, MS-SSIM, and SAM perspectives on the Botswana dataset. (a) PSNRs. (b) MS-SSIMs. (c) SAMs.

spatial structures and perceptual quality. For the SAM, CFSP-Net remains highly competitive and achieves lower spectral distortion than most compared methods. However, PCA+JPEG2000 obtains slightly better SAM values at some bitrate points, which is reasonable because PCA-based spectral decorrelation is particularly effective for preserving the global spectral direction in homogeneous scenes.

Overall, the experimental results demonstrate that CFSP-Net achieves strong and stable compression performance across different hyperspectral datasets. Its advantage is most evi-

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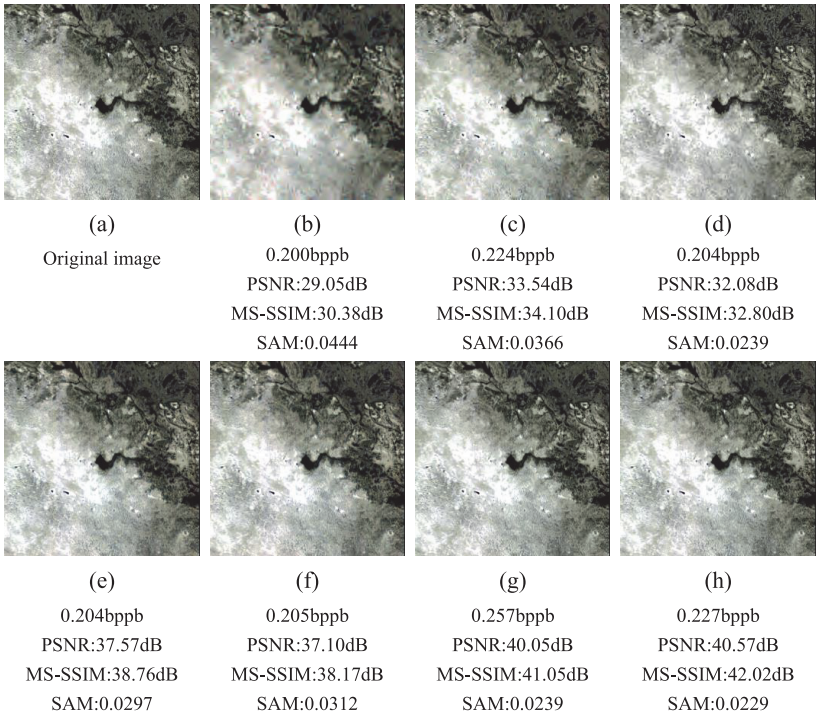


Fig. 13. Visual comparisons at close bit rates on an example patch of the Botswana dataset. Bands 30, 20, and 10 were used to create the false color. (a) Original image. (b) JPEG2000. (c) 3-D-Spiht. (d) PCA+JPEG2000. (e) Scale. (f) Context. (g) FDRCN. (h) Proposed.

TABLE I
DOWNSTREAM CLASSIFICATION RESULTS ON THE PAVIAU DATASET OVER TEN RANDOM SEEDS

Method	bpppb=0.2			bpppb=0.4		
	OA	AA	Kappa	OA	AA	Kappa
Original	0.9300	0.9346	0.9030	0.9300	0.9346	0.9030
JPEG2000	0.8775	0.8833	0.8328	0.8842	0.8871	0.8415
PCA+JPEG2000	0.8965	0.9089	0.8588	0.8999	0.9149	0.8635
3D-SPIHT	0.8828	0.8913	0.8403	0.8990	0.9065	0.8618
Scale Hyperprior	0.8982	0.9194	0.8616	0.9012	0.9223	0.8656
Context2018	0.8963	0.9140	0.8590	0.9008	0.9212	0.8650
FDRCN	0.9044	0.9212	0.8696	0.9004	0.9192	0.8643
CFSP-Net	**0.9110**	**0.9242**	**0.8782**	**0.9055**	**0.9260**	**0.8712**

dent on structurally complex scenes, such as Chikusei and Pavia, where rich textures, fragmented boundaries, and local spatial–spectral variations are difficult to represent using global linear transforms or generic learned codecs. Meanwhile, traditional transform-based methods, especially PCA + JPEG2000, remain competitive in the SAM on datasets with strong global spectral correlation, such as Washington DC Mall and Botswana. Therefore, the performance variation across datasets reflects the interaction between scene characteristics and different compression mechanisms. The proposed CFSP-Net consistently shows superior or highly competitive performance among learning-based methods and provides an effective balance between spatial fidelity, structural similarity, and spectral preservation.

2) Downstream Classification Analysis: To further evaluate whether the reconstructed HSIs preserve task-relevant information for practical remote sensing applications, we conduct a downstream land-cover classification experiment on the Pavia University dataset. The original and reconstructed HSIs obtained by different compression methods are used as the input to the classifier. We adopt the radial basis function support vector machine (RBF-SVM), which is a classical and widely used classifier in HSI classification. For each class, 50 labeled pixels are randomly selected for training, and the remaining labeled pixels are used for testing. To reduce the influence of random sample selection, the experiment is repeated using ten different random seeds. For each random seed, the same training and testing samples are used for all methods to ensure a fair comparison. In Table I, the original HSI is used as a bitrate-independent reference, while the reconstructed HSIs are compared under the corresponding bit rates.

The classification performance is evaluated using overall accuracy (OA), average accuracy (AA), and the Kappa coefficient. OA measures the proportion of correctly classified testing pixels, AA denotes the average of class-wise

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TABLE II
CROSS-SENSOR GENERALIZATION RESULTS ON THE PAVIA UNIVERSITY DATASET

Method	Low bitrate				High bitrate			
	bpppb	PSNR(dB)	MS-SSIM(dB)	SAM(rad)	bpppb	PSNR(dB)	MS-SSIM(dB)	SAM(rad)
JPEG2000	0.1899	28.80	11.78	0.11819	0.4499	32.49	15.98	0.09617
PCA+JPEG2000	0.1938	43.41	27.25	0.03656	0.4562	47.14	30.82	0.02599
3D-SPIHT	0.2128	35.62	19.20	0.08026	0.5040	42.25	25.50	0.05065
Scale Hyperprior	0.1980	35.02	20.56	0.08272	0.5230	35.66	21.44	0.08210
Context2018	0.2120	34.98	20.11	0.08470	0.4980	35.42	21.23	0.08450
FDRCN	0.1912	37.11	19.52	0.08687	0.4465	38.98	21.10	0.07427
CFSP-Net	0.2010	37.23	19.78	0.08245	0.4510	39.43	21.92	0.07130

accuracies, and Kappa evaluates the agreement between the predicted labels and the ground truth while considering chance agreement. These three metrics jointly reflect the overall classification accuracy, class-wise balance, and reliability of the classification results.

As shown in Table I, the proposed method achieves the best classification performance among all compressed reconstruction methods at both bit rates. At around 0.2 bpppb, CFSP-Net obtains an OA of 0.9110, an AA of 0.9242, and a Kappa coefficient of 0.8782. Compared with JPEG2000, PCA+JPEG2000, and 3D-SPIHT, the proposed method improves OA by 3.35, 1.45, and 2.82 percentage points, respectively. Compared with learning-based methods, CFSP-Net also achieves higher OA than Scale Hyperprior, Context2018, and FDRCN by 1.28, 1.47, and 0.66 percentage points, respectively. At around 0.4 bpppb, CFSP-Net also obtains the best compressed-image classification performance, with an OA of 0.9055, an AA of 0.9260, and a Kappa coefficient of 0.8712.

Although the original HSI still provides the highest absolute classification accuracy, the proposed method achieves the closest performance to the original image among all compressed reconstruction methods. These results indicate that CFSP-Net can better preserve task-relevant discriminative information under different compression ratios. Therefore, the reconstructed HSIs produced by the proposed method retain scientific usefulness for downstream land-cover classification, rather than merely achieving favorable reconstruction metrics.

3) Cross-Sensor Generalization Analysis: To further evaluate the generalization ability of the proposed method under sensor-domain shift, we conduct an additional cross-sensor experiment. Specifically, the learning-based models are trained on AVIRIS-source HSI datasets and directly tested on the unseen Pavia University dataset, which has different spectral characteristics and scene contents. During testing, the network structure and model parameters are kept fixed, and no fine-tuning is performed on the target dataset.

The results are reported in Table II. Compared with the learning-based baselines, CFSP-Net achieves the highest PSNR at both low and high bitrates. At around 0.2 bpppb, CFSP-Net obtains a PSNR of 37.23 dB and a SAM of 0.08245. At around 0.45 bpppb, CFSP-Net further improves the PSNR to 39.43 dB and achieves the lowest SAM among the learning-based methods. These results indicate that the proposed coarse-to-fine predictive framework can transfer to unseen HSI data and preserve useful spatial–spectral information under cross-sensor testing.

It is also observed that PCA+JPEG2000 achieves better PSNR, MS-SSIM, and SAM values in this cross-sensor setting. This phenomenon is reasonable because PCA+JPEG2000 performs image-wise orthogonal spectral decorrelation during testing and does not rely on learned source-domain priors. In contrast, deep-learning-based codecs learn statistical priors from the training data, and their performance may degrade when the target dataset has different spectral response, noise characteristics, and spatial texture distribution. Nevertheless, CFSP-Net consistently outperforms the evaluated learning-based compression methods, demonstrating better cross-sensor robustness among neural codecs.

4) Ablation Results: To verify the contribution of each key design in CFSP-Net, we conduct ablation experiments on the Chikusei dataset, which contains complex spatial structures and rich spectral variations. Three variants are constructed: single-stage prediction, w/o SSRB, and w/o residual quantization. The single-stage prediction variant removes the coarse-to-fine prediction process and uses a conventional one-stage prediction network instead. The w/o SSRB variant keeps the coarse-to-fine prediction framework but replaces the spectral–spatial refinement block in the fine prediction stage with a simpler convolutional structure. The w/o residual quantization variant keeps the prediction framework and SSRB, but disables the proposed residual quantization strategy.

As shown in Table III, the full CFSP-Net achieves the best performance at both bitrate levels. Compared with the single-stage prediction variant, the full model improves the PSNR by about 0.7 dB at the low bitrate and about 1.9 dB at the high bitrate. This demonstrates that the coarse-to-fine prediction process is effective, especially when more bits are available to recover fine spatial–spectral details.

The w/o SSRB variant also shows inferior performance compared with the full model. This indicates that the spectral–spatial refinement block plays an important role in the fine prediction stage. Without SSRB, the model has a weaker ability to capture local spatial structures and interband spectral dependencies, leading to lower reconstruction quality and worse spectral fidelity.

In addition, the w/o residual quantization variant performs close to the full model at the low bitrate, but the performance gap becomes larger at the high bitrate. This is because low-bitrate reconstruction is mainly constrained by coarse

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TABLE III
ABLATION STUDY ON THE CHIKUSEI DATASET AT TWO BITRATE RANGES

Variant	Coarse-to-fine	SSRB	Residual quant.	Low bitrate			High bitrate		
				bpppb	PSNR(dB)	SAM(rad)	bpppb	PSNR(dB)	SAM(rad)
Single-stage prediction	×	×	✓	0.194	49.96	0.03325	0.426	53.04	0.03022
w/o SSRB	✓	×	✓	0.205	50.31	0.03248	0.417	53.62	0.02931
w/o residual quant.	✓	✓	×	0.212	50.58	0.03192	0.397	53.85	0.02893
Full CFSP-Net	✓	✓	✓	0.188	**50.72**	**0.03160**	0.410	**54.96**	**0.02730**

TABLE IV
COMPLEXITY COMPARISON OF DIFFERENT COMPRESSION METHODS

Method	Parameters (M)	FLOPs (G)	GPU Mem. (MB)	Enc-time (s)	Dec-time (s)
JPEG2000	–	–	–	1.61	1.12
PCA+JPEG2000	–	–	–	1.78	1.42
3D-SPIHT	–	–	–	4.48	1.53
Scale Hyperprior	20.53	95.97	297.98	0.57	0.61
Context2018	39.25	111.48	398.64	1.37	2.33
FDRCN	16.18	611.61	371.53	0.29	0.17
CFSP-Net	21.43	410.69	376.23	0.38	0.25

structural information, while high-bitrate reconstruction relies more on fine residual representation. Therefore, the proposed residual quantization strategy is particularly beneficial for preserving subtle spatial–spectral details when the quantization step becomes smaller at higher bitrates.

F. Model Complexity Analysis

In addition to R-D performance, computational complexity is also an important factor for evaluating practical HSI compression methods. Therefore, we further compare the proposed CFSP-Net with the competing methods in terms of the number of parameters, FLOPs, GPU memory consumption, encoding time, and decoding time. The results are reported in Table IV. The FLOPs are calculated for one $256 \times 256 \times 50$ HSI patch. For traditional codecs, such as JPEG2000, PCA+JPEG2000, and 3D-SPIHT, the parameters, FLOPs, and GPU memory are not reported because these methods do not rely on trainable neural network models.

As shown in Table IV, CFSP-Net contains 21.43 M parameters, which is close to Scale Hyperprior and much smaller than Context2018. Although the FLOPs of CFSP-Net are higher than those of Scale Hyperprior and Context2018, they are clearly lower than those of FDRCN. This is mainly because CFSP-Net introduces coarse-to-fine prediction and residual refinement modules to enhance spatial–spectral reconstruction quality. These modules increase computational cost, but they also improve the ability of the model to preserve structural details and local spectral variations.

In terms of inference time, CFSP-Net requires 0.38 s for encoding and 0.25 s for decoding, which remains within an acceptable range. Compared with Context2018 and 3D-SPIHT, CFSP-Net achieves faster inference. Although Scale Hyperprior has the lowest computational cost, its reconstruction performance is inferior to CFSP-Net on most datasets. Therefore, CFSP-Net achieves a favorable tradeoff between compression performance and computational complexity.

V. CONCLUSION

In this article, we proposed a novel CFSP-Net to address the persistent challenges of spatial–spectral distortion and high-frequency detail loss in learned HSI compression. Departing from conventional monolithic autoencoders, our framework explicitly decouples the highly complex HSI reconstruction process into two progressive stages. First, a lightweight CNN-based coarse prediction stage extracts low-frequency structural representations to establish a robust spatial–spectral semantic prior. Guided by this prior, a Transformer-based fine prediction stage, equipped with the meticulously designed SSRB, operates at full resolution to progressively restore high-frequency textures and complex interband correlations. Furthermore, the integration of a decoupled residual coding strategy guarantees exceptional ultimate reconstruction fidelity, preventing the blurring of delicate textures under aggressive quantization.

Extensive experiments conducted on multiple benchmark HSI datasets demonstrate that the proposed CFSP-Net achieves strong and highly competitive R-D performance, especially among learning-based methods. Compared with existing traditional codecs and recent deep learning-based compression algorithms, our method significantly improves both spatial perceptual quality and spectral consistency across diverse bitrate scenarios. In future work, we plan to further optimize the computational efficiency of the attention-based modules to facilitate deployment on resource-constrained onboard satellite platforms, and explore the comprehensive impact of our compression framework on downstream analytical tasks, such as spectral unmixing and target detection.

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